

## EAST WEST UNIVERSITY

Department of Computer Science and Engineering  
CSE475 — Machine Learning, Summer 2026

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### Report 1: Exploratory Data Analysis

ds004504 EEG Dataset — Alzheimer's, Frontotemporal Dementia and Healthy Controls

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**Dataset:** OpenNeuro ds004504 – EEG in Alzheimer's Disease,  
Frontotemporal Dementia and Healthy Controls

**Track:** 1 – Graph Neural Network on Table-Style EEG Data

#### Submitted by – Group 03

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This report accompanies the notebook `task1_eda.ipynb`.

## 1. Dataset Overview

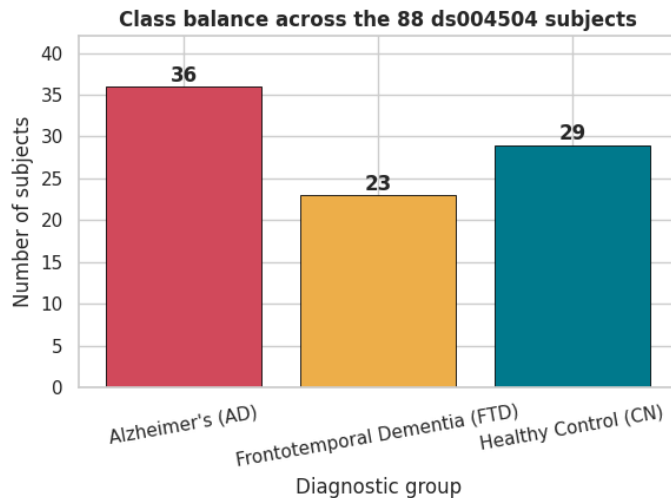
The dataset (OpenNeuro **ds004504**, v1.0.9; Miltiadous et al., 2023) contains resting-state, eyes-closed EEG recordings from **88 subjects**: 19-channel 10–20 montage, 500 Hz sampling rate, mean recording length 13.4 minutes. Each subject is labelled into one of three diagnostic groups.

| Field                       | Value                                  |
|-----------------------------|----------------------------------------|
| Subjects                    | 88                                     |
| EEG channels                | 19 (10–20 system, referential montage) |
| Sampling rate               | 500 Hz                                 |
| Classes                     | 3 (AD, FTD, CN)                        |
| Class counts                | AD = 36, FTD = 23, CN = 29             |
| Recording length            | 13.4 min mean (range 5.1–21.5 min)     |
| Missing values / duplicates | None                                   |

**Table 1:** Dataset summary.

## 2. Class Balance

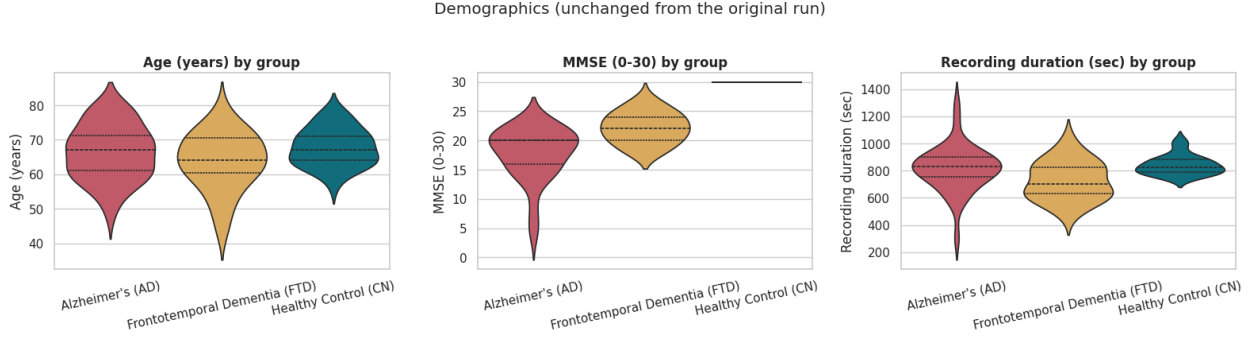
The three diagnostic groups are mildly imbalanced (majority:minority ratio  $\approx 1.57:1$ ), which motivates the use of **macro-F1** rather than plain accuracy as the headline metric for Task 2/3 modelling.



**Figure 1:** Class balance across the 88 subjects: 36 AD, 23 FTD, 29 CN.

## 3. Demographics and Descriptive Statistics

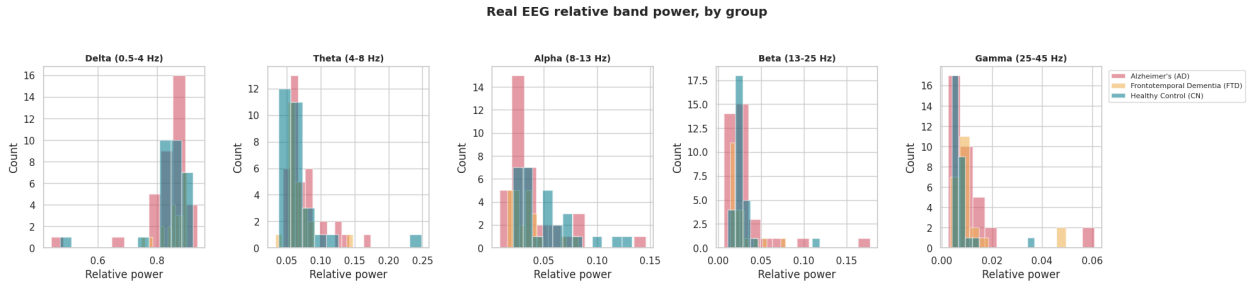
Age, MMSE (cognitive score) and recording duration were examined per group. MMSE is a near-perfect proxy for the diagnostic label (CN subjects score at or near 30), so it is a **leakage risk** and must be excluded from any downstream GNN input features. Whole-scalp relative band power (delta, theta, alpha, beta, gamma) was also summarised: delta dominates (mean  $\approx 0.85$ ) as expected from the 1/f-like nature of EEG power spectra, not a data-quality issue.



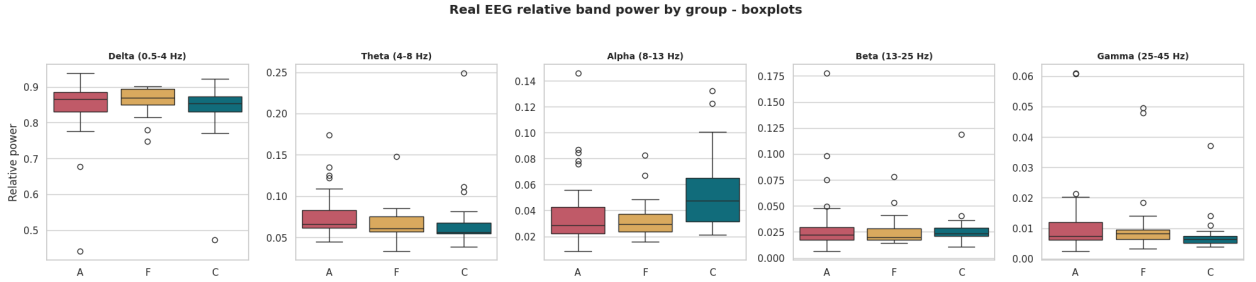
**Figure 2:** Distribution of Age, MMSE, and recording duration by diagnostic group.

#### 4. Relative Band-Power Distributions

The five relative band-power features are plotted by group as histograms and boxplots below.



**Figure 3:** Relative band-power histograms by diagnostic group.



**Figure 4:** Relative band-power boxplots by diagnostic group (A = AD, F = FTD, C = CN).

#### Statistical Test

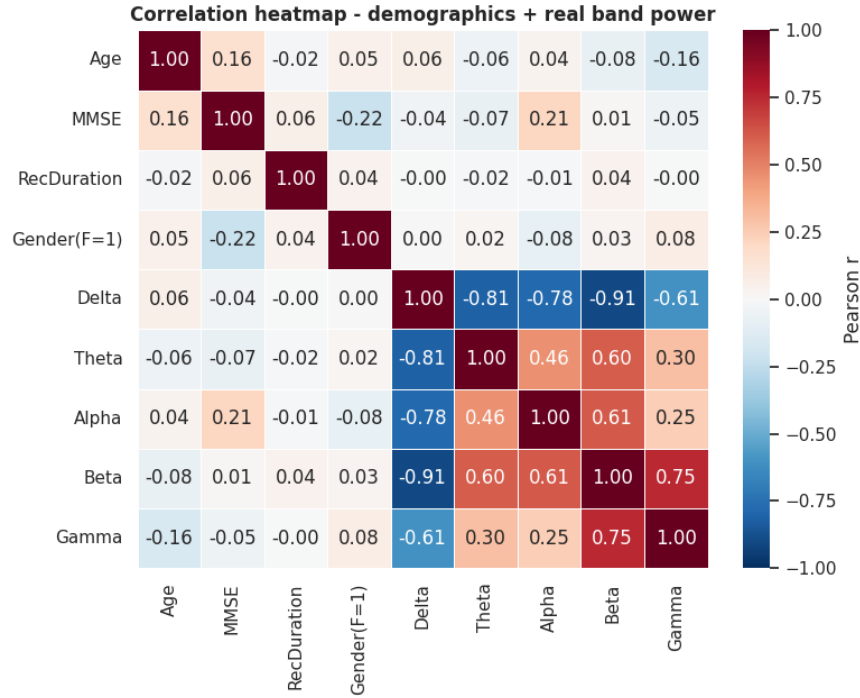
A one-way ANOVA per band (whole-scalp average,  $N=88$ ), with Bonferroni correction for 5 simultaneous tests ( $\alpha = 0.05/5 = 0.01$ ), was used to test for genuine group differences.

| Band             | F-statistic | p-value | Sig. (Bonferroni)     |
|------------------|-------------|---------|-----------------------|
| Delta (0.5–4 Hz) | 0.572       | 0.5666  | No                    |
| Theta (4–8 Hz)   | 0.749       | 0.4758  | No                    |
| Alpha (8–13 Hz)  | 4.018       | 0.0215  | No (uncorrected only) |
| Beta (13–25 Hz)  | 0.378       | 0.6861  | No                    |
| Gamma (25–45 Hz) | 1.268       | 0.2866  | No                    |

**Table 2:** One-way ANOVA results across AD/FTD/CN for each band.

**Finding:** relative alpha power differs across groups at  $p \approx 0.02$  (uncorrected) — CN subjects show the highest mean alpha ( $\approx 0.052$ ), versus AD ( $\approx 0.039$ ) and FTD ( $\approx 0.033$ ). This is directionally consistent with the "EEG slowing" pattern reported in the dementia literature, but does *not* survive Bonferroni correction at  $N=88$ , so it should be treated as suggestive rather than confirmed.

## 5. Correlation Structure

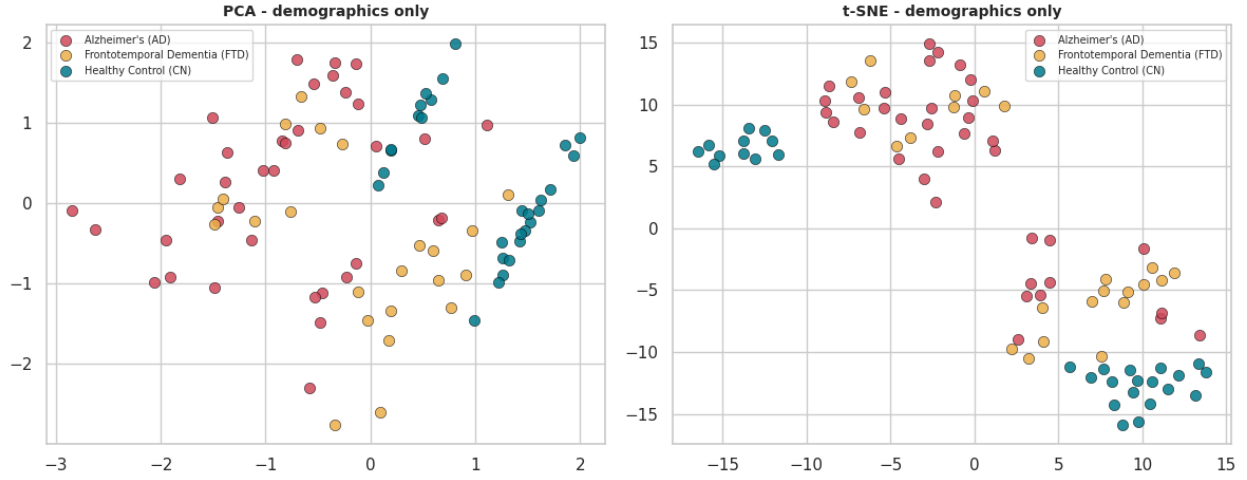


**Figure 5:** Pearson correlation heatmap: demographics and real band-power features.

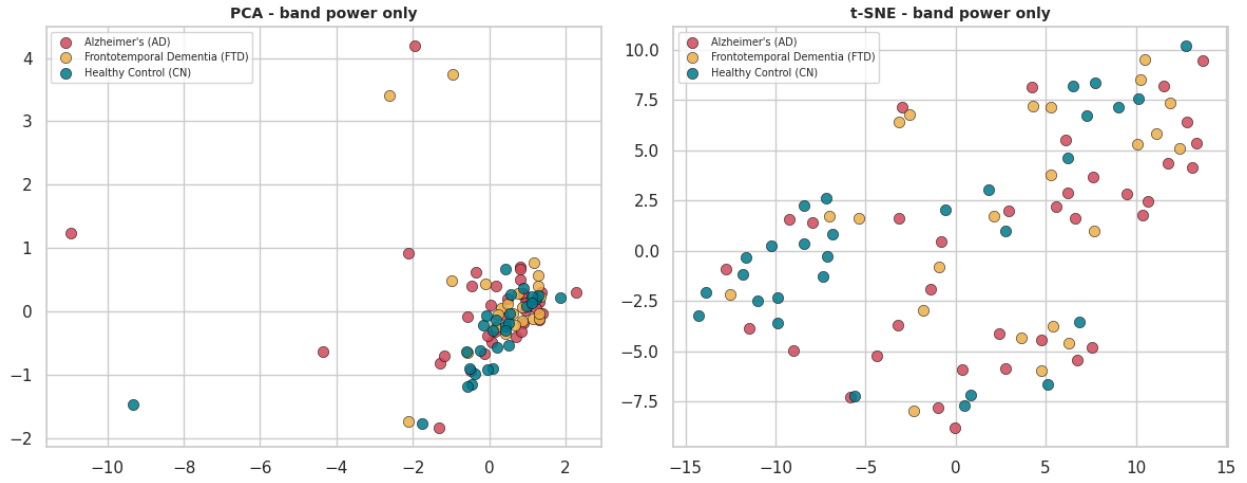
MMSE correlates weakly but visibly with relative alpha (positive), consistent with the ANOVA finding above. As expected for compositional data, delta anti-correlates with the other four bands. No feature pair exceeds the  $|r| \geq 0.70$  redundancy threshold.

## 6. Embedding Analysis (PCA / t-SNE)

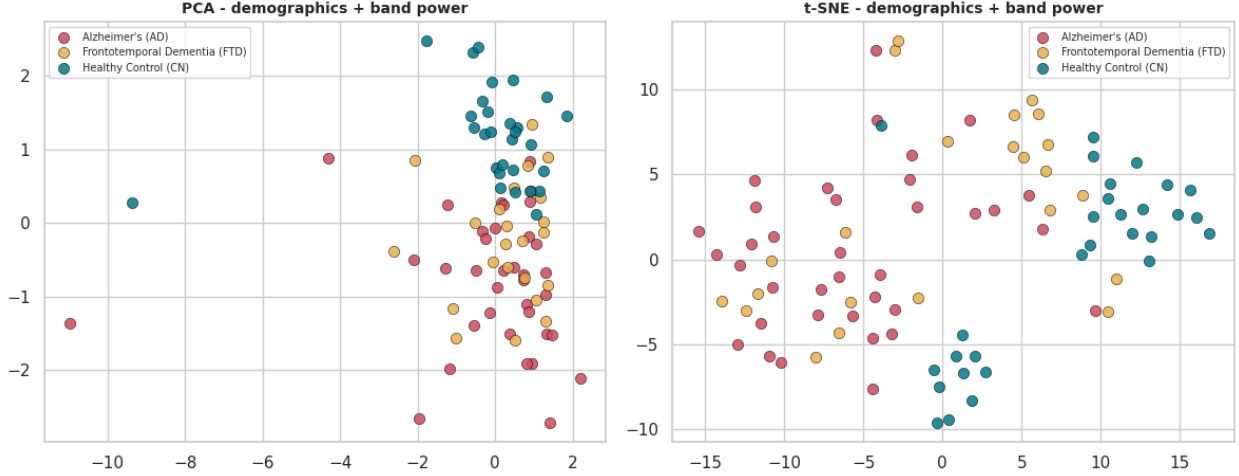
Three feature sets were embedded to visualise class separability: demographics only, band power only, and both combined.



**Figure 6:** PCA and t-SNE using demographic features only (Age, MMSE, Duration, Gender).



**Figure 7:** PCA and t-SNE using whole-scalp band-power features only.



**Figure 8:** PCA and t-SNE using demographics and band-power features combined.

With band power alone, CN separates somewhat from AD/FTD (driven by the alpha finding), but the separation is noisier than the MMSE-driven demographics-only view, and AD/FTD overlap heavily in every version – confirming AD-vs-FTD as the harder sub-problem, and that whole-scalp band power alone is only a weak-to-moderate classification signal.

## 7. Research Gap

1. Most GNN-EEG dementia papers evaluate on private, single-site cohorts and report a single accuracy number rather than macro-F1 with cross-validation.
2. Graph-construction choice (functional-connectivity estimator) is rarely compared against a learned/adaptive graph on the same dataset.
3. AD-vs-FTD discrimination is consistently the hardest sub-problem, confirmed directly in this notebook’s own embeddings.
4. Ternary (3-class) GNN results specifically on ds004504 remain rare.
5. MMSE almost perfectly separates CN from AD/FTD by construction and must be excluded as a model input to avoid leakage.
6. Whole-scalp relative alpha power differs across groups ( $p \approx 0.02$ , uncorrected), but this signal alone is weak – motivating the per-channel, connectivity-graph approach used in Tasks 2 and 3.